MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPES 3WP1 AND 3WP2

The complete requirements for procuring the electron tubes described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Electrostatic deflection and focus

PIN CONNECTIONS AND DIMENSIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	ed	Eb1	Eb2	Rg	Zd	Ehk	Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	Meg	Meg	Vdc	ft
Maximum:	6.9	0, -200	550	1,100	2,750	1.5	1.0	±180	30,000
Minimum:	5.7	---	---	---	1,000	---	---	---	---
TEST CONDITIONS:	6.3	Adjust	---	Focus	1,500	---	---	---	---

GENERAL:

Qualification - Required

(E) denotes changes

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METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Qualification inspection								
1216	Base material insulating quality	Both	Zone 5, or higher	---	---	---	---	---	---
1141	Pressure (implosion)	Both		---	---	---	---	---	---
1002	Barometric pressure, reduced	Both	225 mmHg	---	---	---	---	---	---
5111	Vibration	Both		---	---	Width	---	1	mm
1331	Direct-interelectrode capacitance	Both	g1 to all	---	---	Cg1	---	8.7	pF
			k to all	---	---	Ck	---	5.7	pF
			D1 to D2	---	---	C1D2	---	3.3	pF
			D3 to D4	---	---	C3D4	---	2.0	pF
			D1 to all, except D2	---	---	CD1	---	7.2	pF
			D2 to all, except D1	---	---	CD2	---	7.2	pF
			D3 to all, except D4	---	---	CD3	---	4.8	pF
			D4 to all, except D3	---	---	CD4	---	4.8	pF
5101	Neck and bulb alinement (electrostatic types)	Both		---	---	Dia.	---	1.63	Inch
5216	Cathode illumination	Both		---	---	---	---	---	---
5248	Deflection-factor uniformity	Both		---	---	---	---	2	%
	Quality conformance inspection, part 1								
5106	Bulb, screen, and faceplate quality	Both		---	---	---	---	---	---
5201	Voltage breakdown	Both		---	---	---	---	---	---
5201	Voltage breakdown (electrostatic types)	Both		---	---	---	---	---	---
5206	Gas "cross"	P1	Light = 7 ftL (see note 1)	---	---	---	---	---	---
5206	Gas "cross"	P2	Light = 2 ftL (see note 1)	---	---	---	---	---	---
5221	Light output	P1	See note 1	---	---	Light	7	---	ftL
5221	Light output	P2	See note 1	---	---	Light	2	---	ftL
5231	Spot position (electrostatic deflection)	Both		---	---	---	---	10	mm
5231	Spot displacement (leakage)	Both		---	---	Displ	---	7	mm
5241	Grid cutoff voltage	Both		---	---	Eco	-45	-75	Vdc
5251	Grid No. 1 leakage current	Both		---	---	---	---	---	---
5251	Anode No. 2 leakage current	Both		---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 1 -Continued								
---	Useful scan	Both	Measured from tube face center; Focused trace						
			1D2 scan	---	---	---	2.50	---	Inch
			3D4 scan	---	---	---	2.25	---	Inch
(E) 5103	Pattern distortion	Both	See note 2	---	---	---	---	---	---
	Quality conformance inspection, part 2								
1301	Heater current	Both		---	---	If	540	660	mA
5201	Electrode current (cathode)	P1	Light = 7 ftL	---	---	Ik	---	1,000	μAdc
5201	Electrode current (cathode)	P2	Light = 2 ftL	---	---	Ik	---	1,000	μAdc
5201	Electrode current (anode No. 1)	Both	Ec1 = 0	---	---	Ib1	-15	10	μAdc
5101	Base alinement (electrostatic types)	Both	+1D2; pin No. 3	---	---	---	---	---	---
5101	Neck and base alinement (electrostatic types)	Both		---	---	---	---	---	---
5101	Angle between traces	Both		---	---	---	89	91	Degrees
5216	Stray light emission (conventional types)	Both	Eb2 = 2,750 Vdc	---	---	---	---	---	---
5223	Modulation	P1	Light = 7 ftL	---	---	ΔEc	---	50	Vdc
5223	Modulation	P2	Light = 2 ftL	---	---	ΔEc	---	45	Vdc
5226	Line width A (electrostatic deflection)	P1	Light = 7 ftL	---	---	Width	---	0.65	mm
5226	Line width A (electrostatic deflection)	P2	Light = 2 ftL	---	---	Width	---	0.65	mm
5226	Line width B (electrostatic deflection)	P1	Light = 7 ftL	---	---	Width	---	0.75	mm
5226	Line width B (electrostatic deflection)	P2	Light = 2 ftL	---	---	Width	---	0.75	mm
5246	Focusing voltage at modulation condition	Both		---	---	Eb1	247	465	Vdc
5248	Deflection factor	Both	1D2	---	---	DF	62	76	Vdc/in.
5248	Deflection factor	Both	3D4	---	---	DF	43	52	Vdc/in.
5251	Heater-cathode leakage current	Both		---	---	---	---	---	---
(E) 1111	Base pin solder depth	Both		---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 2 -Continued								
1101	Secureness of base, cap, or insert	Both		---	---	---	---	---	---
1105	Permanence of marking	Both		---	---	---	---	---	---
	Quality conformance inspection, part 3								
---	Life test	Both	Group C; t = 500 hours minimum; Eb2 = 2,750 Vdc	---	---	---	---	---	---
		P1	Light = 7 ftL	---	---	---	---	---	---
		P2	Light = 2 ftL	---	---	---	---	---	---
---	Life-test end points	Both	Line width A	---	---	Width	---	0.65	mm
			Line width B	---	---	Width	---	0.75	mm
			Heater-cathode leakage current	---	---	---	---	---	---
			Grid No. 1 leakage current	---	---	---	---	---	---
			Grid No. 2 leakage current	---	---	---	---	---	---
			Anode No. 1 leakage current	---	---	---	---	---	---
			Stray light emission	---	---	---	---	---	---
		P1	Light = 4.5 ftL	---	---	---	---	---	---
			Modulation	---	---	ΔEc	---	50	Vdc
		P2	Light = 1.5 ftL	---	---	---	---	---	---
			Modulation	---	---	ΔEc	---	45	Vdc

NOTES:

1. This test to be performed at the conclusion of the holding period.
2. With a raster pattern adjusted so that its widest point just touches the sides of a 2.05-inch (52.07 mm) square, no point on the raster sides shall be within an inscribed 1.95-inch (49.53 mm) square.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Air Force - 11, 17
DSA - ES

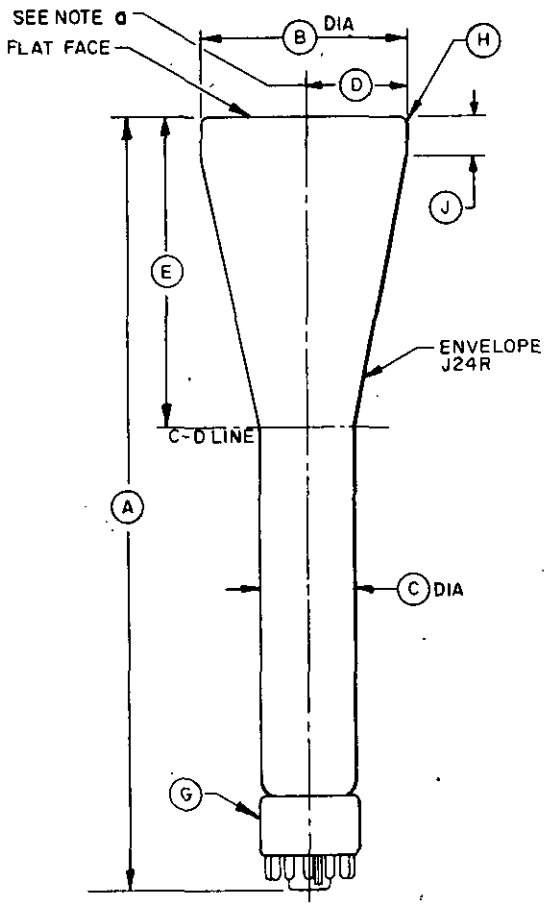
User activities:
Army - MU, WC
Navy - AS, MC, CG, SH

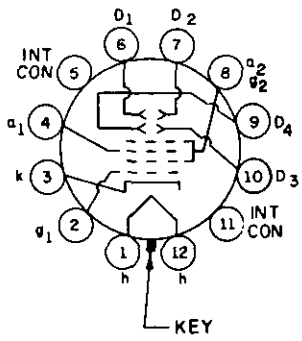
Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2530)

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BASING DIAGRAM

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Qualification inspection		
B	2.938 (74.63)	3.063 (77.80)
G	Base: B12-43 or B10-75	
Quality conformance inspection, part 2		
A	11.375(288.93)	11.625(295.28)
C	1.313 (33.35)	1.438 (36.53)
D	1.375 (34.93)R	
Reference dimensions (see note b)		
E	4.563(115.90)	
H	0.063 (1.60)R	
J	0.563 (14.30)	

NOTES:
a. Useful screen radius.
b. Reference dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube types 3WP1 and 3WP2.

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